

Exploring the impact of resource limitations on gene network reconstruction

Tyler Quarton, Taek Kang, Eduardo D. Sontag, and Leonidas Bleris

Abstract—The characterization of biological networks via mathematical models often involves cycles of experimental perturbations and measurements, followed by the use of a network inference method. Here we study an engineered genetic circuit, introduced in a recent paper by the authors, and report additional analysis and interpretation. Using this synthetic network as a benchmark, we find that the application of the modular response analysis (MRA) network inference method leads to the discovery of a hidden, nontrivial “ghost” regulatory edge, which was not explicitly engineered into the network. Importantly, this result is not evident from direct inspection of the experimental measurements and global response coefficients. To probe the global to local conversion in MRA, we use conditionally randomized global response matrices to obtain distributions of local response coefficients and demonstrate that sign changes are numerically possible. Additionally, using simulations of a cascade network in a biochemical setting which does not take into account resource limitations, we show that MRA cannot return “ghost” edges, which points to the impact of the cellular milieu and in particular the use of shared resources. Taking resource availability into account during reverse engineering may allow for closer approximation of the cellular environment and points to a potential opportunity for network characterization strategies.

I. INTRODUCTION

Reverse engineering of gene regulatory networks is a fundamental area of research in systems and synthetic biology [1], [2], [3], [4]. The availability of massive sets of gene expression data has shifted the focus from describing interactions among several genes to comprehensive analysis at the network level [5], [6], [7], [8]. As the scope of analysis expands to a larger system, the dynamics of common cellular resources must be factored into the analysis as well. The abundance of factors that govern the central dogma are generally understood to be monitored and regulated throughout the cell cycle [9], [10], [11], and consequently have a profound impact on the activity of the cellular network [12], [13].

Mathematical frameworks that enable the analysis of resource derived network disturbances have been established. Specifically, network effects caused by molecular competition were derived using Metabolic Control Analysis [14], metabolic constraints were established and explored using Flux Balance Analysis [15], design rules to avoid

intranetwork cross couplings were outlined through enzymatic competition considerations [16], and queuing models were proposed to explore the effects of ribosome bottlenecking [17], [18]. In a recent work, Qian et al. developed a framework which accounts for the effects arising from the sharing of RNA polymerase and ribosomes in gene networks specifically [19].

We use as a benchmark a synthetic three-node cascade motif that is orthogonal to the cellular milieu but operates using shared resources [1]. We subject our network to iterations of perturbation, measurement and reverse engineering [2]. The synthetic network, which consists of three distinct nodes with their own fluorescent reporters, features an indirect regulation between the input and the output node (i.e. X activating Y, Y activating Z). After perturbation-based network inference, we show that a weak direct regulatory relationship between the input (X) and output node (Z) is uncovered, in addition to the intended connections. Here we study the origin of this ghost edge uncovered during network inference. This conference paper discusses experimental results and numerical explorations. A journal paper with more details as well as more complete theoretical developments is in preparation.

II. BENCHMARK SYNTHETIC CIRCUIT

The model synthetic network consists of three independent nodes in the cascade motif, where node X activates node Y, which, in turn, activates node Z (Fig. 1). Each node features a distinct reporter fluorescent protein and controls its respective downstream node via inducible transcriptional activation. Perturbation of each node in the system was made possible by introducing small interfering RNA (siRNA) that directly inhibit translation of the fluorescent protein as well as the co-expressed transactivator proteins of the node.

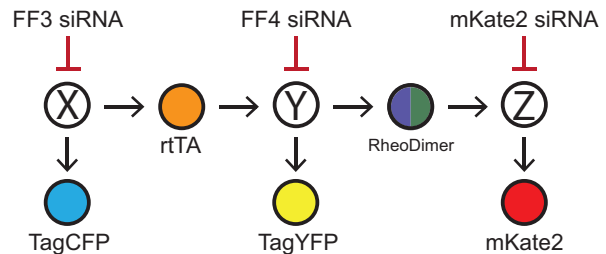


Fig. 1. Graphical representation of the synthetic benchmark circuit. Each node produces fluorescent reporter protein as well as the transcriptional activator protein to its downstream node. Perturbation of each node made possible through siRNA [1].

The entire synthetic network was transiently delivered to human embryonic kidney (HEK293) cells (Fig. 2). Node X is

Tyler Quarton and Taek Kang are with the Department of Bioengineering of the University of Texas at Dallas, U.S.A.

Eduardo D. Sontag is with the Department of Mathematics of Rutgers University, NJ, U.S.A., Email address: eduardo.sontag@rutgers.edu

Leonidas Bleris is with the Department of Bioengineering of the University of Texas at Dallas, U.S.A. Email address: bleris@utdallas.edu

This work was funded by NSF grants DMS-13611355, CBET-1351354, and AFOSR grant FA9550-14-1-0060.

constitutively active, doxycycline enables node Y activation by node X, and ponasterone A mediates activation of node Z by node Y. As there are multiple transcripts that must be targeted to disrupt nodes X and Y, perturbations of these nodes are performed by synthetic siRNAs whose target sequences are located in 3' UTR of all transcripts produced [5].

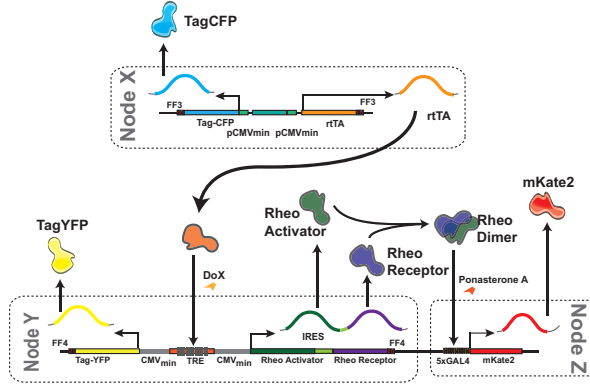


Fig. 2. Components of the synthetic cascade network. Each node produces a fluorescent reporter protein as well as a transactivator protein for its downstream target activation, with the exception of node Z. Ubiquitous disruption of nodes X and Y are achieved via siRNA targets in the 3' UTR of the transcripts produced [1].

The regulatory interactions of the synthetic circuit are mediated by chemical ligands doxycycline and ponasterone A (Fig. 2). Activity of node X, represented by TagCFP, is constitutive and produces transactivator rtTA. In combination with doxycycline, rtTA activates node Y to produce TagYFP and both components of RheoSwitch transactivator. Finally, node Z is activated when an ecdysone receptor agonist such as ponasterone A binds to the RheoSwitch dimer complex to enable Gal4 domain activity and production of mKate2 fluorescent protein.

To confirm the architecture of the synthetic network, we performed two separate validation tests [1]. First, the circuit plasmids are transfected transiently and the activity of each output protein is measured 48 hours post-transfection after various combinations of the ligands that mediate node-to-node interactions (Fig. 3)[1].

Since multiple components are produced in nodes X and Y, we opted for perturbation using synthetic siRNAs and placed their targets at the 3' UTR of every transcript produced at each node. For node X, siRNA FF3 was used, and for node Y, FF4 siRNA. For node Z, we used a custom siRNA that exclusively targets the fluorescent protein's transcript [1]. We validated this node-specific perturbation through titration of each siRNA. The siRNAs are co-transfected with the circuit, and as before the outputs are measured after 48 hours via flow cytometry (Fig. 4).

III. NETWORK RECONSTRUCTION

A. Modular response analysis

When it comes to perturbation-based network reconstruction methods, accurately distinguishing direct interactions

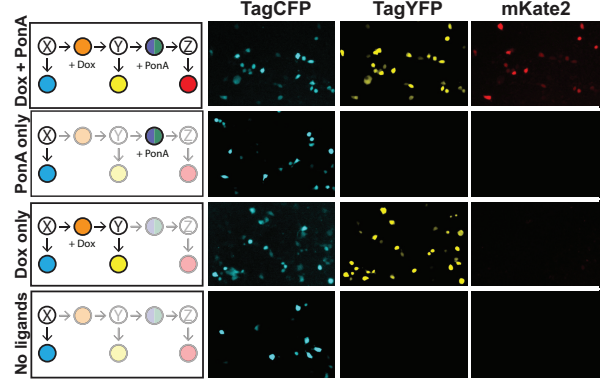


Fig. 3. Confirmation of the regulatory connections in the synthetic cascade network. Network activity was measured via fluorescence microscopy under all combinations of the inducer ligands that activate the regulatory links of the network [1].

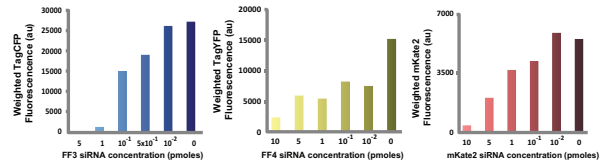


Fig. 4. Individual node response following perturbation with siRNA that target each node. Each node of the synthetic cascade network is perturbed via RNA interference. To measure the efficacy of each siRNA, titrations were performed [1].

from indirect interactions is one of the most important and complex tasks. Perturbation to a single node may propagate through the network and results in changes at a global scale, which renders inference of pairwise interactions particularly challenging. Over the years, there have been several attempts to address this issue [20], [21]. One such method is modular response analysis (MRA), which uses linear algebra techniques to derive local, pairwise interaction coefficients from global response data [22] [23].

The MRA method has previously shown to be successful in characterization of the canonical Raf/Mek/Erk MAPK network pathway [24]. The procedure for reverse engineering of our synthetic cascade network using MRA is as follows: first, we capture the steady states of the system after siRNA perturbation for each node at a specified concentration, which we use to calculate the global response coefficients. After constructing a matrix consisting of the response coefficients, we then use this matrix to calculate the local response coefficients [23].

B. Review of the MRA method

We consider a system of differential equations of the following general form:

$$\dot{x}_i = F_i(x_1, \dots, x_N, p_i), \quad i = 1, \dots, N$$

where each F_i is continuously differentiable. The variables $x_i, i = 1, \dots, N$ denote concentrations of species, and each $p_i, i = 1, \dots, N$ is a parameter. Note that the i th parameter only affects directly the i th equation.

The MRA method amounts precisely to the following procedure. We are interested in finding the direct dependencies of each variable x_i on each other variable x_j . More precisely, we assume that $\xi = (\xi_1, \dots, \xi_N)^T$ is a steady state of the system corresponding to a nominal parameter $\pi = (\pi_1, \dots, \pi_N)^T$, that is,

$$F_i(\xi_1, \dots, \xi_N, \pi_i) = 0, \quad i = 1, \dots, N$$

and we want to estimate the gradients ∇F_i evaluated at (ξ, π) , $i = 1, \dots, N$. We assume that there is a smooth curve $x(p)$ of steady states, where p is some neighborhood of the nominal parameter π , and $x(\pi) = \xi$ and denote by

$$\frac{dx_k}{dp_j}$$

the global change in the k th coordinate, x_k , of the steady state when p_j is perturbed. Taking derivatives with respect to p_j , $j \neq i$ in $F_i(x_1(p), \dots, x_N(p), p_i) \equiv 0$, we obtain:

$$\sum_{k=1}^N \frac{\partial F_i}{\partial x_k} \frac{dx_k}{dp_j} = 0 \quad \forall j \neq i. \quad (1)$$

Thus we have the orthogonality equations

$$\nabla F_i \cdot P_j = 0, \quad i = 1, \dots, N \quad \forall j \neq i,$$

where

$$P_j = \left(\frac{dx_1}{dp_j}, \frac{dx_2}{dp_j}, \dots, \frac{dx_N}{dp_j} \right).$$

From steady-state information, we can never find the gradients of the F_i 's except up to a constant multiple. So we attempt to estimate the normalized *local response coefficients*, defined as follows:

$$r_{ij} := -\frac{\partial F_i}{\partial x_j} / \frac{\partial F_i}{\partial x_i}$$

for $i, j \in \{1, \dots, N\}$, assuming that $\frac{\partial F_i}{\partial x_i} \neq 0$ (typically negative, hence the negative sign). Note that, by definition, $r_{ii} = -1$ for all i . The matrix $\mathbf{r}$ consisting of the r_{ij} 's is called the *local response matrix*. Observe that (1), together with the definition of r_{ij} , gives that, for all i and all $j \neq i$,

$$\frac{dx_i}{dp_j} = \sum_{k \neq i} r_{ik} \frac{dx_k}{dp_j}. \quad (2)$$

We also write, purely formally,

$$\frac{\partial x_i}{\partial x_j} := -\frac{\partial F_i}{\partial x_j} / \frac{\partial F_i}{\partial x_i} = r_{ij}. \quad (3)$$

Intuitively, we interpret each $\frac{\partial x_i}{\partial x_j}$, $i \neq j$, as a perturbation in the activity x_i of species i with respect to infinitesimal perturbations in the activity x_j of species j under the idealized assumption that one could keep the activities of all other species clamped. These coefficients are all understood as evaluated at $x = \xi$ and $p = \pi$.

Fix any i . Assuming that F_i is continuously differentiable and $\frac{\partial F_i}{\partial x_i} \neq 0$ at $x = \xi$, $p = \pi$, the Implicit Function Theorem says that there is a function

$$x_i = f_i(x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_N, p)$$

defined in a neighborhood of (ξ, π) , with the property that, as an equation on X_i ,

$$F_i(x_1, x_2, \dots, x_{i-1}, X_i, x_{i+1}, \dots, x_N, p) = 0$$

has a unique solution in a neighborhood of (ξ, π) , given by $X_i = f_i(x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_N, p)$, and furthermore in this neighborhood

$$\frac{\partial F_i}{\partial x_k} = -\frac{\partial F_i}{\partial x_i} \frac{\partial f_i}{\partial x_k} \quad \forall k \neq i. \quad (4)$$

These formulas therefore hold also when specialized to $x = x(p)$ for parameters p near the nominal parameter π , and in particular at $x = \xi$.

Thus,

$$\frac{\partial f_i}{\partial x_k} = -\frac{\partial F_i}{\partial x_j} / \frac{\partial F_i}{\partial x_i} = r_{ij} = \frac{\partial x_i}{\partial x_k} \quad \forall k \neq i.$$

Now, substituting (4) in (1) we have

$$\frac{\partial F_i}{\partial x_i} \frac{dx_i}{dp_j} - \sum_{k \neq i} \frac{\partial F_i}{\partial x_i} \frac{\partial f_i}{\partial x_k} \frac{dx_k}{dp_j} = 0 \quad \forall j \neq i. \quad (5)$$

Let us define the *global response coefficients* as follows:

$$R_{ij} = \frac{dx_i}{dx_j} := \frac{dx_i}{dp_j} / \frac{dx_j}{dp_j} \quad \forall i, j.$$

Note that, by definition, $R_{ii} = 1$ for all i . The matrix $\mathbf{R}$ consisting of the R_{ij} 's is called the *global response matrix*. Observe that

$$\frac{dx_k}{dp_j} = \frac{dx_k}{dx_j} \frac{dx_j}{dp_j} \quad \forall k, j.$$

Substituting these expressions in (5), and then factoring out $\frac{\partial F_i}{\partial x_i} \frac{dx_j}{dp_j}$ (generically assumed nonzero), we arrive at:

$$\frac{dx_i}{dx_j} = \sum_{k \neq i} \frac{\partial f_i}{\partial x_k} \frac{dx_k}{dx_j} \quad \forall i \neq j,$$

which can also be written when using (3) as:

$$\frac{dx_i}{dx_j} = \sum_{k \neq i} \frac{\partial x_i}{\partial x_k} \frac{dx_k}{dx_j} \quad \forall i \neq j, \quad (6)$$

or, equivalently in terms of the $\mathbf{r}$ and $\mathbf{R}$ notations:

$$R_{ij} = \sum_{k \neq i} r_{ik} R_{kj} \quad \forall i \neq j. \quad (7)$$

This is the basic MRA equation that relates local and global coefficients. Assuming that the global coefficients R_{ij} are estimated from perturbation experiments, equation (7) allows one to solve for the local coefficients, which indicate direct interactions [23].

C. Experimental perturbation and measurements

Using the flow cytometry data gathered after perturbation, we proceeded with the network inference using MRA (Fig. 5) [1]. The global response coefficients (GRC) were calculated using bootstrapped mean of the gated fluorescent population, which are in turn used to calculate local response coefficients (LRC). To obtain a reasonable confidence interval associated with each LRC, we performed the calculation cycle 2,000 times and use 95% confidence interval of the resulting LRC distributions for final results. In other words, we only consider recovered LRC as significant if the 95% confidence interval does not intersect zero.

While we are able to recover positive, non-zero LRC in edges representing interaction from node X to node Y and node Y to node Z, we unexpectedly discovered an apparent direct interaction between node X to node Z. This reaction coefficient, though considered negligible due to its confidence interval, is significantly higher in magnitude than other unintended edges such as node Y to node X and node Z to node Y. Moreover, the same pattern was observed (Fig. 5) when another set of siRNAs was used to perform systematic perturbation at a different level (i.e. low and high perturbation).

To summarize, MRA applied on the benchmark architecture yields a nontrivial “ghost” edge. In the next section, we study the properties of the global to local response coefficient conversion in MRA using randomized matrices and numerical simulations of cascade architectures.

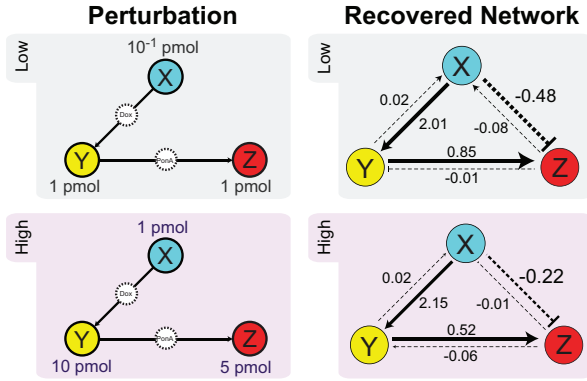


Fig. 5. Graphical representation of the perturbation properties. Systematic perturbation to the network is performed twice, each with a different set of siRNA concentrations. Each siRNA concentrations are categorized as either “Low” or “High” to indicate the relative strength of suppression. Interaction coefficients for all regulatory links were then calculated using modular response analysis using mean of fluorescence [1].

IV. RANDOMIZED CASCADES AND SIMULATIONS

Firstly, to numerically explore the effects of processing global measurements to local responses, we performed MRA analysis on 100,000 randomized 3-node cascade networks. Using the central fractional difference to quantify perturbation results [25], given generally as

$$\frac{2(P_{\text{pert}} - P_{\text{unpert}})}{(P_{\text{pert}} + P_{\text{unpert}})}, \quad (8)$$

where P is the quantity being perturbed, we constructed corresponding global response matrices which represents all possible perturbations and after applying MRA we obtained distributions of local response coefficients. Secondly, we simulated the genetic circuit behavior and response to perturbation using a representative ODE model in MATLAB SimBiology.

A. Randomized Cascades

To study the distribution of all possible local response coefficients derived from perturbation of a cascade architecture we apply the MRA using conditionally randomized global response matrices. We restricted our analysis to direct 3 node cascades and include only the expected perturbation responses. Specifically, when X is perturbed, we expect X, Y, and Z to decrease, when Y is perturbed, we expect Y and Z to decrease, and when Z is perturbed, we expect only Z to decrease. By inspection, the maximum value of the central fractional difference (8) of two positive measurements upon negative perturbation is 0 and the minimum is -2. Therefore, all possible direct cascade central fractional differences are contained within the range $[-2, 0]$. Thus, the global response elements of our randomize 3-node direct cascades are designed as follows

$$\mathbf{R}_{ij} = \begin{cases} \sim U([-2, 0]) & i \geq j \\ 0 & i < j \end{cases}, \quad (9)$$

where $\sim U([-2, 0])$ indicates a random value from a uniform distribution ranging from -2 to 0. After generating our cascade perturbation results, we subjected each global matrix to MRA analysis and recovered their local responses. By looking at the probability density function corresponding to the histogram of the recovered local response coefficients, we confirm that the responses r_{yx} and r_{zy} are strictly positive and, interestingly, r_{zy} has a non-zero probability of being negative (Fig. 6). Clearly, MRA has the numerical capacity to produce a negative edge from nodes X to Z.

B. Perturbation Simulations

A genetic cascade comprised of nodes that represent DNA, RNA, and protein species were modeled using MATLAB SimBiology using previously published kinetic parameters [5]. To emulate various strengths of siRNA perturbation on nodes X, Y, and Z, we gradually increase the mRNA degradation of each respective node. The cascade’s dynamics were then simulated (Fig. 7) and steady-state values were obtained. Using the the central fractional difference (8) as before, we calculate the global response matrices for each perturbation strength. After subjecting these global responses to MRA, we observe that in the case of our in-silico transcriptional cascade, we recover positive values for responses r_{yx} and r_{yz} while r_{zx} is zero for all perturbation strengths (Fig. 8). The fact that a negative edge was not recovered after perturbing a genetic cascade in silico, yet is numerically possible, indicates that the ghost edge may be a consequence of factors that are not explicitly modeled, including the assumed abundant shared resources.

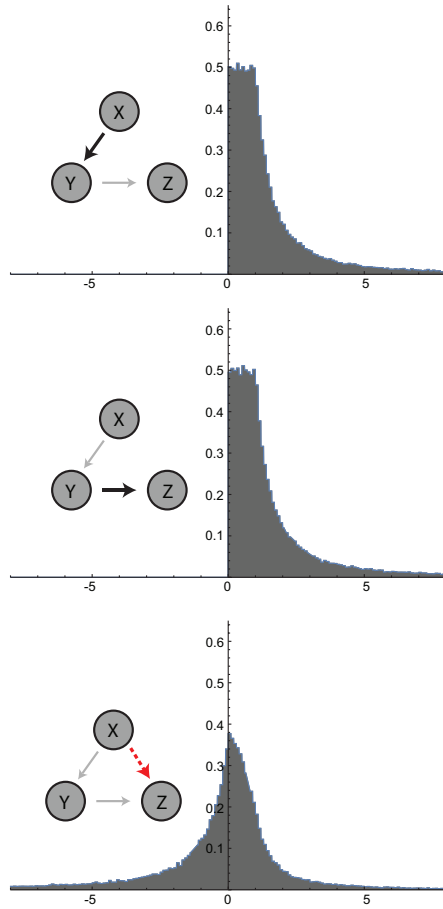


Fig. 6. Probability density functions of local response coefficients obtained from random cascades. Only the probability density function for r_{zx} (bottom) includes both positive and negative coefficients.

V. CONCLUSIONS

Synthetic biological networks as benchmarks [1], [26], [2], [27], [28] have provided a unique opportunity to study biological reverse engineering methods. The reconstruction results of our circuit via MRA uncover a “ghost” regulation edge not readily apparent from experimental measurements and accordingly, requires further investigation of the cellular environment complexities. The emergent edge of the resource constrained cascade implicates the possible existence of a naturally formed genetic “dark network”, where collections of hidden edges affect complex network dynamics in unexplored and nuanced ways from the shadows of operational context. A necessary first step in uncovering the secrets of the dark genetic network would be to rigorously validate the existence of other hidden edges through modified inference methods and classify their interactions within visible genetic networks and among themselves.

REFERENCES

- [1] T. Kang, R. Moore, Y. Li, E. Sontag, and L. Bleris, “Discriminating direct and indirect connectivities in biological networks,” *Proceedings of the National Academy of Sciences*, vol. 112, no. 41, pp. 12 893–12 898, October 13 2015.

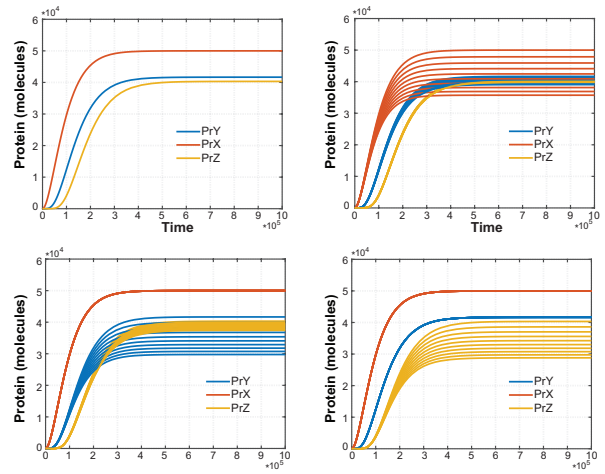


Fig. 7. SimBiology perturbation simulations. The model cascade was serially perturbed at 9 different levels of increasing mRNA degradation at each respective node. Cascade dynamics for node X, Y, and Z were simulated for each perturbation case.

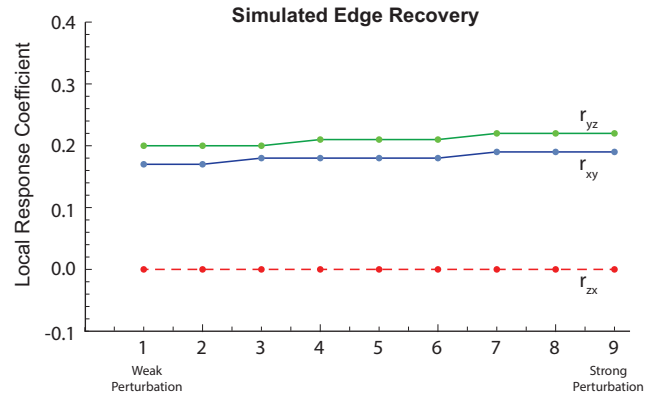


Fig. 8. Local response coefficients of simulated data. Responses r_{yx} and r_{xy} were found to be positive while r_{zx} was found to be 0 for all perturbation strengths.

- [2] T. Kang, J. T. White, Z. Xie, Y. Benenson, E. Sontag, and L. Bleris, “Reverse engineering validation using a benchmark synthetic gene circuit in human cells,” *ACS synthetic biology*, vol. 2, no. 5, pp. 255–262, May 17 2013.
- [3] T. S. Gardner, D. di Bernardo, D. Lorenz, and J. J. Collins, “Inferring genetic networks and identifying compound mode of action via expression profiling,” *Science*, vol. 301, no. 5629, pp. 102–105, 2003. [Online]. Available: <http://science.sciencemag.org/content/301/5629/102>
- [4] D. Marbach, R. J. Prill, T. Schaffter, C. Mattiussi, D. Floreano, and G. Stolovitzky, “Revealing strengths and weaknesses of methods for gene network inference,” *Proceedings of the National Academy of Sciences*, vol. 107, no. 14, pp. 6286–6291, 2010. [Online]. Available: <http://www.pnas.org/content/107/14/6286.abstract>
- [5] L. Bleris, Z. Xie, D. Glass, A. Adadey, E. Sontag, and Y. Benenson, “Synthetic incoherent feedforward circuits show adaptation to the amount of their genetic template,” *Molecular systems biology*, vol. 7, p. 519, Aug 2 2011.
- [6] V. Shimoga, J. T. White, Y. Li, E. Sontag, and L. Bleris, “Synthetic mammalian transgene negative autoregulation,” *Molecular systems biology*, vol. 9, p. 670, Jun 4 2013.
- [7] B. Hayete, T. S. Gardner, and J. J. Collins, “Size matters: network inference tackles the genome scale,” *Molecular Systems Biology*, vol. 3, no. 1, 2007. [Online]. Available: <http://www.mol-sys.org/content/3/1/1>

<http://msb.embopress.org/content/3/1/77>

- [8] K. Basso, A. A. Margolin, G. Stolovitzky, U. Klein, R. Dalla-Favera, and A. Califano, "Reverse engineering of regulatory networks in human b cells," *Nature genetics*, vol. 37, no. 4, pp. 382–390, 2005.
- [9] N. Sonenberg and A. G. Hinnebusch, "Regulation of translation initiation in eukaryotes: Mechanisms and biological targets," *Cell*, vol. 136, no. 4, pp. 731–745, 2/20 2009.
- [10] J. Houseley and D. Tollervey, "The many pathways of rna degradation," *Cell*, vol. 136, no. 4, pp. 763–776, 2/20 2009.
- [11] F. Spitz and E. E. Furlong, "Transcription factors: from enhancer binding to developmental control," *Nature Reviews Genetics*, vol. 13, no. 9, pp. 613–626, Sep 2012.
- [12] S. Cardinale and A. P. Arkin, "Contextualizing context for synthetic biology—identifying causes of failure of synthetic biological systems," *Biotechnology journal*, vol. 7, no. 7, pp. 856–866, 2012.
- [13] O. Borkowski, F. Ceroni, G.-B. Stan, and T. Ellis, "Overloaded and stressed: whole-cell considerations for bacterial synthetic biology," *Current Opinion in Microbiology*, vol. 33, pp. 123 – 130, 2016.
- [14] D. De Vos, F. J. Bruggeman, H. V. Westerhoff, and B. M. Bakker, "How molecular competition influences fluxes in gene expression networks," *PLoS ONE*, vol. 6, no. 12, pp. 1–15, 12 2011.
- [15] N. Abedpour and M. Kollmann, "Resource constrained flux balance analysis predicts selective pressure on the global structure of metabolic networks," *BMC Systems Biology*, vol. 9, no. 1, pp. 1–13, 2015. [Online]. Available: <http://dx.doi.org/10.1186/s12918-015-0232-5>
- [16] Y. Rondelez, "Competition for catalytic resources alters biological network dynamics," *Physical review letters*, vol. 108, no. 1, p. 018102, 2012.
- [17] W. H. Mather, J. Hasty, L. S. Tsimring, and R. J. Williams, "Translational cross talk in gene networks," *Biophysical journal*, vol. 104, no. 11, pp. 2564–2572, 2013.
- [18] A. Raveh, M. Margaliot, E. D. Sontag, and T. Tuller, "A model for competition for ribosomes in the cell," *Journal of The Royal Society Interface*, vol. 13, no. 116, p. 20151062, 2016.
- [19] Y. Qian and D. Del Vecchio, "Effective interaction graphs arising from resource limitations in gene networks," in *2015 American Control Conference (ACC)*. IEEE, 2015, pp. 4417–4423.
- [20] S. Feizi, D. Marbach, M. Medard, and M. Kellis, "Network deconvolution as a general method to distinguish direct dependencies in networks," *Nature biotechnology*, vol. 31, no. 8, pp. 726–733, Aug 2013.
- [21] B. Kholodenko, M. B. Yaffe, and W. Kolch, "Computational approaches for analyzing information flow in biological networks," *Science signaling*, vol. 5, no. 220, p. re1, Apr 17 2012.
- [22] B. N. Kholodenko, A. Kiyatkin, F. J. Bruggeman, E. Sontag, H. V. Westerhoff, and J. B. Hoek, "Untangling the wires: a strategy to trace functional interactions in signaling and gene networks," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 99, no. 20, pp. 12 841–12 846, Oct 1 2002.
- [23] E. Sontag, "Network reconstruction based on steady-state data," *Essays in Biochemistry*, vol. 45, pp. 161–176, 2008.
- [24] S. D. M. Santos, P. J. Verveer, and P. I. H. Bastiaens, "Growth factor-induced mapk network topology shapes erk response determining pc-12 cell fate," *Nature cell biology*, vol. 9, no. 3, pp. 324–330, 2007.
- [25] B. N. Kholodenko, A. Kiyatkin, F. J. Bruggeman, E. Sontag, H. V. Westerhoff, and J. B. Hoek, "Untangling the wires: A strategy to trace functional interactions in signaling and gene networks," *Proceedings of the National Academy of Sciences*, vol. 99, no. 20, pp. 12 841–12 846, 2002. [Online]. Available: <http://www.pnas.org/content/99/20/12841.abstract>
- [26] I. Cantone, L. Marucci, F. Iorio, M. A. Ricci, V. Belcastro, M. Bansal, S. Santini, M. Di Bernardo, D. Di Bernardo, and M. P. Cosma, "A yeast synthetic network for in vivo assessment of reverse-engineering and modeling approaches," *Cell*, vol. 137, no. 1, pp. 172–181, 2009.
- [27] D. Marbach, T. Schaffter, C. Mattiussi, and D. Floreano, "Generating realistic in silico gene networks for performance assessment of reverse engineering methods," *Journal of computational biology*, vol. 16, no. 2, pp. 229–239, 2009.
- [28] F. Menolascina, M. Di Bernardo, and D. Di Bernardo, "Analysis, design and implementation of a novel scheme for in-vivo control of synthetic gene regulatory networks," *Automatica*, vol. 47, no. 6, pp. 1265–1270, 2011.